

# Package ‘betaSandwich’

February 28, 2026

**Title** Robust Confidence Intervals for Standardized Regression Coefficients

**Version** 1.0.8.9000

**Description** Generates robust confidence intervals for standardized regression coefficients using heteroskedasticity-consistent standard errors for models fitted by `lm()` as described in Dudgeon (2017) <[doi:10.1007/s11336-017-9563-z](https://doi.org/10.1007/s11336-017-9563-z)>. The package can also be used to generate confidence intervals for R-squared, adjusted R-squared, and differences of standardized regression coefficients. A description of the package and code examples are presented in Pesigan, Sun, and Cheung (2023) <[doi:10.1080/00273171.2023.2201277](https://doi.org/10.1080/00273171.2023.2201277)>.

**URL** <https://github.com/jeksterslab/betaSandwich>,  
<https://jeksterslab.github.io/betaSandwich/>

**BugReports** <https://github.com/jeksterslab/betaSandwich/issues>

**License** MIT + file LICENSE

**Encoding** UTF-8

**LazyData** true

**Roxygen** list(markdown = TRUE)

**Depends** R (>= 3.5.0)

**Suggests** knitr, rmarkdown, testthat, betaDelta, betaMC, betaNB

**RoxygenNote** 7.3.3.9000

**NeedsCompilation** no

**Author** Ivan Jacob Agaloos Pesigan [aut, cre, cph] (ORCID:  
<<https://orcid.org/0000-0003-4818-8420>>)

**Maintainer** Ivan Jacob Agaloos Pesigan <[r.jeksterslab@gmail.com](mailto:r.jeksterslab@gmail.com)>

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|         |                                                                                                                                                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| BetaADF | <i>Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach</i> |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

---

**Description**

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach

**Usage**

```
BetaADF(object, alpha = c(0.05, 0.01, 0.001))
```

**Arguments**

- object            Object of class lm.
- alpha            Numeric vector. Significance level  $\alpha$ .

**Details**

Note that while the calculation in `BetaADF()` is different from `betaDelta::BetaDelta()` with `type = "adf"`, the results are numerically equivalent. `BetaADF()` is appropriate when sample sizes are moderate to large ( $n > 250$ ). `BetaHC()` is recommended in most situations.

**Value**

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm\_process** Processed `lm` object.

**gamma\_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma\_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, 37(1), 62–83. doi:[10.1111/j.20448317.1984.tb00789.x](https://doi.org/10.1111/j.20448317.1984.tb00789.x)

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, 82(4), 928–951. doi:[10.1007/s113360179563z](https://doi.org/10.1007/s113360179563z)

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). `betaDelta` and `betaSandwich`: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:[10.1080/00273171.2023.2201277](https://doi.org/10.1080/00273171.2023.2201277)

**See Also**

Other Beta Sandwich Functions: [BetaHC\(\)](#), [BetaN\(\)](#), [DiffBetaSandwich\(\)](#), [RSqBetaSandwich\(\)](#)

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaADF(object)
# Methods -----
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

|        |                                                                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BetaHC | <i>Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach</i> |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach

## Usage

```
BetaHC(
  object,
  type = "hc3",
  alpha = c(0.05, 0.01, 0.001),
  g1 = 1,
  g2 = 1.5,
  k = 0.7
)
```

## Arguments

|                     |                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class <code>lm</code> .                                                                            |
| <code>type</code>   | Character string. Correction type. Possible values are "hc0", "hc1", "hc2", "hc3", "hc4", "hc4m", and "hc5". |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ .                                                                |
| <code>g1</code>     | Numeric. g1 value for type = "hc4m".                                                                         |
| <code>g2</code>     | Numeric. g2 value for type = "hc4m".                                                                         |
| <code>k</code>      | Numeric. Constant k for type = "hc5" $0 \leq k \leq 1$ .                                                     |

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm\_process** Processed `lm` object.

**gamma\_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma\_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**References**

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, 82(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

**See Also**

Other Beta Sandwich Functions: [BetaADF\(\)](#), [BetaN\(\)](#), [DiffBetaSandwich\(\)](#), [RSqBetaSandwich\(\)](#)

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
# Methods -----
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

BetaN

*Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality*

---

**Description**

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality

**Usage**

```
BetaN(object, alpha = c(0.05, 0.01, 0.001))
```

**Arguments**

|        |                                               |
|--------|-----------------------------------------------|
| object | Object of class <code>lm</code> .             |
| alpha  | Numeric vector. Significance level $\alpha$ . |

## Details

Note that while the calculation in `BetaN()` is different from `betaDelta::BetaDelta()` with `type = "mvn"`, the results are numerically equivalent. `BetaN()` assumes multivariate normality. `BetaHC()` is recommended in most situations.

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm\_process** Processed `lm` object.

**gamma\_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma\_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, 82(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). `betaDelta` and `betaSandwich`: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `DiffBetaSandwich()`, `RSqBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaN(object)
# Methods -----
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

|                   |                                       |
|-------------------|---------------------------------------|
| coef.betasandwich | <i>Standardized Regression Slopes</i> |
|-------------------|---------------------------------------|

---

**Description**

Standardized Regression Slopes

**Usage**

```
## S3 method for class 'betasandwich'  
coef(object, ...)
```

**Arguments**

|        |                               |
|--------|-------------------------------|
| object | Object of class betasandwich. |
| ...    | additional arguments.         |

**Value**

Returns a vector of standardized regression slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)  
std <- BetaHC(object)  
coef(std)
```

---

|                       |                                                      |
|-----------------------|------------------------------------------------------|
| coef.diffbetasandwich | <i>Differences of Standardized Regression Slopes</i> |
|-----------------------|------------------------------------------------------|

---

**Description**

Differences of Standardized Regression Slopes

**Usage**

```
## S3 method for class 'diffbetasandwich'  
coef(object, ...)
```

**Arguments**

object            Object of class diffbetasandwich.  
 ...              additional arguments.

**Value**

Returns a vector of differences of standardized regression slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
coef(diff)
```

---

coef.rsqbetasandwich    *Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

---

**Description**

Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

**Usage**

```
## S3 method for class 'rsqbetasandwich'
coef(object, ...)
```

**Arguments**

object            Object of class rsqbetasandwich.  
 ...              additional arguments.

**Value**

Returns a vector of multiple correlation coefficients (R-squared and adjusted R-squared)

**Author(s)**

Ivan Jacob Agaloos Pesigan



**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
coef(rsq)
```

---

confint.betasandwich    *Confidence Intervals for Standardized Regression Slopes*

---

**Description**

Confidence Intervals for Standardized Regression Slopes

**Usage**

```
## S3 method for class 'betasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

**Arguments**

|        |                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class betasandwich.                                                                                                                                         |
| parm   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level  | the confidence level required.                                                                                                                                        |
| ...    | additional arguments.                                                                                                                                                 |

**Value**

Returns a matrix of confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
confint(std, level = 0.95)
```

---

`confint.diffbetasandwich`*Confidence Intervals for Differences of Standardized Regression Slopes*

---

**Description**

Confidence Intervals for Differences of Standardized Regression Slopes

**Usage**

```
## S3 method for class 'diffbetasandwich'  
confint(object, parm = NULL, level = 0.95, ...)
```

**Arguments**

|                     |                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class <code>diffbetasandwich</code> .                                                                                                                       |
| <code>parm</code>   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| <code>level</code>  | the confidence level required.                                                                                                                                        |
| <code>...</code>    | additional arguments.                                                                                                                                                 |

**Value**

Returns a matrix of confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)  
std <- BetaHC(object)  
diff <- DiffBetaSandwich(std)  
confint(diff, level = 0.95)
```

---

`confint.rsqbetasandwich`*Confidence Intervals for Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

---

**Description**

Confidence Intervals for Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

**Usage**

```
## S3 method for class 'rsqbetasandwich'  
confint(object, parm = NULL, level = 0.95, ...)
```

**Arguments**

|                     |                                                                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class <code>rsqbetasandwich</code> .                                                                                                                        |
| <code>parm</code>   | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| <code>level</code>  | the confidence level required.                                                                                                                                        |
| <code>...</code>    | additional arguments.                                                                                                                                                 |

**Value**

Returns a matrix of confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)  
std <- BetaHC(object)  
rsq <- RSqBetaSandwich(std)  
confint(rsq, level = 0.95)
```

---

|                  |                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------|
| DiffBetaSandwich | <i>Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix</i> |
|------------------|-----------------------------------------------------------------------------------------------------|

---

## Description

Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix

## Usage

```
DiffBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

|        |                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| object | Object of class betasandwich, that is, the output of the <a href="#">BetaHC()</a> , <a href="#">BetaN()</a> , or <a href="#">BetaADF()</a> functions. |
| alpha  | Numeric vector. Significance level $\alpha$ .                                                                                                         |

## Value

Returns an object of class diffbetasandwich which is a list with the following elements:

**call** Function call.

**fit** The argument object.

**args** Function arguments.

**vcov** Sampling covariance matrix of differences of standardized slopes.

**est** Vector of differences of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: [BetaADF\(\)](#), [BetaHC\(\)](#), [BetaN\(\)](#), [RSqBetaSandwich\(\)](#)

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
# Methods -----
print(diff)
summary(diff)
coef(diff)
vcov(diff)
confint(diff, level = 0.95)
```

nas1982

*1982 National Academy of Sciences Doctoral Programs Data***Description**

1982 National Academy of Sciences Doctoral Programs Data

**Usage**

nas1982

**Format**

Ratings of 46 doctoral programs in psychology in the USA with the following variables:

**QUALITY** Program quality ratings.**NFACUL** Number of faculty members in the program.**NGRADS** Number of program graduates.**PCTSUPP** Percentage of program graduates who received support.**PCTGRT** Percent of faculty members holding research grants.**NARTIC** Number of published articles attributed to program faculty member.**PCTPUB** Percent of faculty with one or more published article.**References**

National Research Council. (1982). *An assessment of research-doctorate programs in the United States: Social and behavioral sciences*. doi:10.17226/9781. Reproduced with permission from the National Academy of Sciences, Courtesy of the National Academies Press, Washington, D.C.

print.betasandwich

*Print Method for an Object of Class betasandwich***Description**

Print Method for an Object of Class betasandwich

**Usage**

```
## S3 method for class 'betasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

**Arguments**

|                     |                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>      | Object of class <code>betasandwich</code> .                                                                                               |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ . If <code>alpha = NULL</code> , use the argument <code>alpha</code> used in <code>x</code> . |
| <code>digits</code> | Digits to print.                                                                                                                          |
| <code>...</code>    | additional arguments.                                                                                                                     |

**Value**

Prints a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
print(std)
```

---

```
print.diffbetasandwich
```

*Print Method for an Object of Class diffbetasandwich*

---

**Description**

Print Method for an Object of Class `diffbetasandwich`

**Usage**

```
## S3 method for class 'diffbetasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

**Arguments**

|                     |                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>      | Object of class <code>diffbetasandwich</code> .                                                                                           |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ . If <code>alpha = NULL</code> , use the argument <code>alpha</code> used in <code>x</code> . |
| <code>digits</code> | Digits to print.                                                                                                                          |
| <code>...</code>    | additional arguments.                                                                                                                     |

**Value**

Prints a matrix of differences of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
print(diff)
```

---

print.rsqbetasandwich *Print Method for an Object of Class rsqbetasandwich*

---

**Description**

Print Method for an Object of Class rsqbetasandwich

**Usage**

```
## S3 method for class 'rsqbetasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

**Arguments**

|                     |                                                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>      | Object of class rsqbetasandwich.                                                                                                          |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ . If <code>alpha = NULL</code> , use the argument <code>alpha</code> used in <code>x</code> . |
| <code>digits</code> | Digits to print.                                                                                                                          |
| <code>...</code>    | additional arguments.                                                                                                                     |

**Value**

Prints a matrix of multiple correlation coefficients (R-squared and adjusted R-squared), standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
print(rsq)
```

---

|                 |                                                                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| RSqBetaSandwich | <i>Estimate Multiple Correlation Coefficients (R-squared and adjusted R-squared) and the Corresponding Sampling Covariance Matrix</i> |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|

---

## Description

Estimate Multiple Correlation Coefficients (R-squared and adjusted R-squared) and the Corresponding Sampling Covariance Matrix

## Usage

```
RSqBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

|               |                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>object</b> | Object of class betasandwich, that is, the output of the <a href="#">BetaHC()</a> , <a href="#">BetaN()</a> , or <a href="#">BetaADF()</a> functions. |
| <b>alpha</b>  | Numeric vector. Significance level $\alpha$ .                                                                                                         |

## Value

Returns an object of class rsqbetasandwich which is a list with the following elements:

**call** Function call.

**fit** The argument object.

**args** Function arguments.

**vcov** Sampling covariance matrix of multiple correlation coefficients (R-squared and adjusted R-squared).

**est** Vector of multiple correlation coefficients (R-squared and adjusted R-squared).

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: [BetaADF\(\)](#), [BetaHC\(\)](#), [BetaN\(\)](#), [DiffBetaSandwich\(\)](#)



**Examples**

```

object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
# Methods -----
print(rsq)
summary(rsq)
coef(rsq)
vcov(rsq)
confint(rsq, level = 0.95)

```

---

summary.betasandwich    *Summary Method for an Object of Class betasandwich*

---

**Description**

Summary Method for an Object of Class betasandwich

**Usage**

```

## S3 method for class 'betasandwich'
summary(object, alpha = NULL, digits = 4, ...)

```

**Arguments**

|        |                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------|
| object | Object of class betasandwich.                                                                         |
| alpha  | Numeric vector. Significance level $\alpha$ . If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print.                                                                                      |
| ...    | additional arguments.                                                                                 |

**Value**

Returns a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```

object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
summary(std)

```

---

summary.diffbetasandwich

*Summary Method for an Object of Class diffbetasandwich*


---

## Description

Summary Method for an Object of Class diffbetasandwich

## Usage

```
## S3 method for class 'diffbetasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

## Arguments

|        |                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------|
| object | Object of class diffbetasandwich.                                                                     |
| alpha  | Numeric vector. Significance level $\alpha$ . If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print.                                                                                      |
| ...    | additional arguments.                                                                                 |

## Value

Returns a matrix of differences of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
summary(diff)
```

---

`summary.rsqbetasandwich`*Summary Method for an Object of Class rsqbetasandwich*

---

## Description

Summary Method for an Object of Class rsqbetasandwich

## Usage

```
## S3 method for class 'rsqbetasandwich'  
summary(object, alpha = NULL, digits = 4, ...)
```

## Arguments

|                     |                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>object</code> | Object of class rsqbetasandwich.                                                                                                               |
| <code>alpha</code>  | Numeric vector. Significance level $\alpha$ . If <code>alpha = NULL</code> , use the argument <code>alpha</code> used in <code>object</code> . |
| <code>digits</code> | Digits to print.                                                                                                                               |
| <code>...</code>    | additional arguments.                                                                                                                          |

## Value

Returns a matrix of multiple correlation coefficients (R-squared and adjusted R-squared), standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)  
std <- BetaHC(object)  
rsq <- RSqBetaSandwich(std)  
summary(rsq)
```

---

|                   |                                                                         |
|-------------------|-------------------------------------------------------------------------|
| vcov.betasandwich | <i>Sampling Covariance Matrix of the Standardized Regression Slopes</i> |
|-------------------|-------------------------------------------------------------------------|

---

**Description**

Sampling Covariance Matrix of the Standardized Regression Slopes

**Usage**

```
## S3 method for class 'betasandwich'
vcov(object, ...)
```

**Arguments**

|        |                               |
|--------|-------------------------------|
| object | Object of class betasandwich. |
| ...    | additional arguments.         |

**Value**

Returns a matrix of the variance-covariance matrix of standardized slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
vcov(std)
```

---

|                       |                                                                                    |
|-----------------------|------------------------------------------------------------------------------------|
| vcov.diffbetasandwich | <i>Sampling Covariance Matrix of Differences of Standardized Regression Slopes</i> |
|-----------------------|------------------------------------------------------------------------------------|

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**Description**

Sampling Covariance Matrix of Differences of Standardized Regression Slopes

**Usage**

```
## S3 method for class 'diffbetasandwich'
vcov(object, ...)
```

**Arguments**

object            Object of class `diffbetasandwich`.  
 ...              additional arguments.

**Value**

Returns a matrix of the variance-covariance matrix of differences of standardized regression slopes.

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
vcov(diff)
```

---

`vcov.rsqbetasandwich`    *Sampling Covariance Matrix of Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

---

**Description**

Sampling Covariance Matrix of Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

**Usage**

```
## S3 method for class 'rsqbetasandwich'
vcov(object, ...)
```

**Arguments**

object            Object of class `rsqbetasandwich`.  
 ...              additional arguments.

**Value**

Returns a matrix of the variance-covariance matrix of multiple correlation coefficients (R-squared and adjusted R-squared).

**Author(s)**

Ivan Jacob Agaloos Pesigan

**Examples**

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
vcov(rsq)
```

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